

Supplementary Information for “Validation-Aligned Coreset Selection for Budgeted Few-Shot Classification”

Supplementary Note 1: claim boundary

The main article reports a bounded claim. VACS-R is the strongest evaluated training-only deployable protocol in the five-dataset low-budget 3-nearest-neighbour benchmark, but the experiments do not establish broad state-of-the-art performance. The supplementary tables below document the paired fast-protocol audit, ablations, non-deployable hindsight-static reference, validation-size diagnostics, selection costs, and frozen embedding boundary checks.

Supplementary Table 1: paired low-budget fast-protocol stability

This table reports VACS-F minus herding differences over matched low-budget units.

Slice	<i>n</i>	Δ Acc.	95% CI	W/T/L	Sign <i>p</i>
Pooled	80	+0.02	[-1.38,+1.20]	28/41/11	0.009
20NG	16	+4.54	[+2.43,+7.07]	16/0/0	< .001
Breast cancer	16	+0.84	[+0.37,+1.32]	7/9/0	0.016
Digits	16	+0.19	[-0.12,+0.53]	3/12/1	0.625
Iris	16	-4.31	[-9.86,-0.14]	2/9/5	0.453
Wine	16	-1.16	[-2.20,-0.35]	0/11/5	0.062

Supplementary Table 2: ablation diagnostics

The ablation table separates the validation wrapper from individual selector candidates and from the final rebuild convention.

Variant	Acc.	Δ	Macro-F1	Selected
Full VACS	70.6	+0.00	70.2	11.4
No final rebuild	72.0	+1.34	71.7	11.4
No K-means candidate	70.4	-0.21	69.8	11.4
No MARC candidate	70.7	+0.08	70.3	11.4
No random candidate	70.6	-0.05	70.3	11.4

Supplementary Table 3: non-deployable dataset-hindsight static audit

The per-dataset static reference is selected with hindsight from the low-budget result table, and is therefore diagnostic rather than deployable.

Slice	Static ref.	VACS	Static	Δ [95% CI]	W/T/L
Pooled per-dataset-best	Per-dataset	70.6	72.3	-1.64 [-2.98,-0.61]	8/53/19
20NG	Random	9.7	12.1	-2.49 [-5.33,-0.03]	4/7/5
Breast cancer	K-means	93.4	93.5	-0.15 [-0.37,+0.00]	0/14/2
Digits	K-means	82.3	82.4	-0.12 [-0.50,+0.19]	2/12/2
iris	Herding	76.4	80.7	-4.31 [-9.86,-0.14]	2/9/5
Wine	Herding	91.3	92.5	-1.16 [-2.20,-0.35]	0/11/5

Supplementary Table 4: validation-size reliability

This diagnostic repeats the internal validation-ranking analysis at 10%, 20%, 25%, and 33% validation fractions.

Val. frac.	Units	VACS	Gain	Top-1	ρ	Regret	Severe
0.10	64	69.1	+0.94	60.9	0.81	1.58	12.5
0.20	64	68.3	+0.10	46.9	0.77	2.42	18.8
0.25	64	69.3	+1.13	60.9	0.81	1.38	9.4
0.33	64	69.0	+0.86	59.4	0.80	1.66	14.1

Supplementary Table 5: selection cost

The cost diagnostic reports selection-only time for herding and validation-plus-rebuild time for VACS-F over the matched low-budget decisions.

Slice	n	Herding	VACS val.	Rebuild	Total
Pooled	80	10.475	79.820	7.390	87.210
20NG	16	10.250	75.500	6.859	82.359
Breast cancer	16	0.038	0.969	0.107	1.076
Digits	16	0.137	2.011	0.336	2.347
Iris	16	0.018	0.667	0.038	0.705
Wine	16	0.032	0.673	0.050	0.723

Supplementary Table 6: frozen transformer probes

The frozen transformer probes are boundary evidence. They do not support a broad modern text-embedding superiority claim.

Probe	Dim.	VACS	Best static	Gain
DeBERTa-v3-small	768	9.8	Herding 9.9	-0.17
SmolLM2-360M, maxlen 64	960	31.5	MARC 31.4	+0.03
SmolLM2-360M, maxlen 128	960	37.4	Herding 37.5	-0.12
Qwen2.5-0.5B	896	40.1	Herding 40.2	-0.11

Supplementary Tables 7 and 8: frozen image-embedding probes

The image probes widen benchmark coverage but tie the strongest static selector in the tested one-seed, one-example-per-class settings.

Dataset	VACS	Best static	Gain	Backbone
Fashion-MNIST	75.6	Herding 75.6	+0.00	convnext_tiny
Pooled	75.6	Herding 75.6	+0.00	convnext_tiny

Dataset	VACS	Best static	Gain	Backbone
CIFAR-10	64.7	Herding 64.7	+0.00	resnet18
Pooled	64.7	Herding 64.7	+0.00	resnet18

Supplementary Table 9: full 20-class sparse-text stress boundary

The compact full-20-class TF-IDF stress check is negative for VACS relative to static random and is kept as a claim boundary.

Task	Classes	Units	VACS	Best static	Gain	Top VACS choice
20 Newsgroups (20-class)	20	16	9.7	Random 12.1	-2.49	Boundary 9/16

Supplementary Note 2: reproducibility files

The project package includes the Python scripts used to regenerate the primary result manifest, all tables included in the manuscript and supplementary information, the workflow and diagnostic

1 figures, and the consistency checks that compare manuscript claims with generated result files. The
2 main experiment driver is `experiments/run_vacs_experiments.py`. Covertypes audits,
3 repeated-validation audits, validation-size diagnostics, and figure generation are implemented as
4 separate companion scripts for auditability.